WORKED SOLUTION

STEP I 2015, Question 6

Pure • Vectors in Geometry

STEP I 2015, Question 6

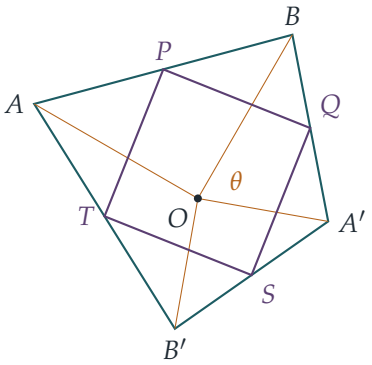
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q6

The key idea

With O as origin, the conditions $OA = OB$ and $\angle AOB = 90°$ say that B is A turned through a right angle about O, and likewise B' is A' turned the same way. Write R for that quarter-turn. Then each side of the midpoint quadrilateral turns out to be the previous side with R applied, and that is exactly what makes a square. Once the side is known in terms of $\mathbf{a}$ and $\mathbf{a}'$, the area is a cosine-rule calculation.

Solution



The midpoints form a square

Take O as origin and write $\mathbf{a}, \mathbf{b}, \mathbf{a}', \mathbf{b}'$ for the position vectors of A, B, A', B'. Let R be rotation through $90°$ clockwise about O. It is linear, it preserves lengths, and it turns every non-zero vector through a right angle.

Because the vertices go round clockwise and O lies inside the quadrilateral, going from OA to OB is a clockwise turn of $90°$. Since $OA = OB$, this gives $\mathbf{b} = R\mathbf{a}$. In the same way $\mathbf{b}' = R\mathbf{a}'$.

Let P, Q, S, T be the midpoints of AB, BA', $A'B'$, $B'A$:

$$\mathbf{p} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}), \quad \mathbf{q} = \tfrac{1}{2}(\mathbf{b}+\mathbf{a}'), \quad \mathbf{s} = \tfrac{1}{2}(\mathbf{a}'+\mathbf{b}'), \quad \mathbf{t} = \tfrac{1}{2}(\mathbf{b}'+\mathbf{a}).$$

Now take the four sides in turn. Let $\mathbf{u} = \overrightarrow{PQ}$:

$$\begin{aligned}
\overrightarrow{PQ} &= \tfrac{1}{2}(\mathbf{a}'-\mathbf{a}) = \mathbf{u},\\
\overrightarrow{QS} &= \tfrac{1}{2}(\mathbf{b}'-\mathbf{b}) = \tfrac{1}{2}R(\mathbf{a}'-\mathbf{a}) = R\mathbf{u},\\
\overrightarrow{ST} &= \tfrac{1}{2}(\mathbf{a}-\mathbf{a}') = -\mathbf{u},\\
\overrightarrow{TP} &= \tfrac{1}{2}(\mathbf{b}-\mathbf{b}') = -R\mathbf{u}.
\end{aligned}$$

Here $\mathbf{u} \neq \mathbf{0}$, because $A \neq A'$. So PQ and ST are equal and opposite, which makes $PQST$ a parallelogram. Its adjacent side QS is $\mathbf{u}$ turned through a right angle with no change in length. A parallelogram with equal, perpendicular adjacent sides is a square, so $PQST$ is a square, as required.

Ask yourself Why must the quarter-turn taking A to B be the same one that takes A' to B'?

Both pairs come one after the other in the clockwise labelling, and O is inside, so in each case the turn from the first ray to the second is clockwise. If one turn were clockwise and the other anticlockwise, the step from $\overrightarrow{QS}$ to $R\mathbf{u}$ would break down, and in general the figure would not be a square.

Maximising the area

Let $OA = OB = p$ and $OA' = OB' = q$, both fixed, and let $\theta = \angle BOA'$. The four angles at O add up to 360°, and two of them are right angles, so $\angle B'OA = 180^\circ - \theta$. As O is inside, both θ and $180^\circ - \theta$ lie strictly between 0° and 180°, so $0 < \theta < 180^\circ$.

The square has side $|\mathbf{u}| = \frac{1}{2}AA'$, so its area is $\frac{1}{4}AA'^2$. Turning from OA through OB to OA', the angle between $\mathbf{a}$ and $\mathbf{a}'$ is $90^\circ + \theta$, and $\cos(90^\circ + \theta) = -\sin\theta$. By the cosine rule (or by expanding $|\mathbf{a}' - \mathbf{a}|^2$),

$$AA'^2 = p^2 + q^2 - 2pq\cos(90^\circ+\theta) = p^2 + q^2 + 2pq\sin\theta,$$

so

$$\text{area} = \tfrac{1}{4}\left(p^2 + q^2 + 2pq\sin\theta\right).$$

With p and q fixed, this is largest when $\sin\theta$ is largest. On $0 < \theta < 180^\circ$ that happens exactly at

$$\angle BOA' = \theta = 90^\circ,$$

and the maximum area is $\frac{1}{4}(p+q)^2$. In this case all four angles at O are right angles, so A' lies on AO extended beyond O, and the side of the square is $\frac{1}{2}(p+q)$.

Checks

- Take $p = 2$, $q = 3$. The area formula gives $\frac{1}{4}(13 + 12\sin\theta)$, which is 3.25 near $\theta = 0$, rises to $6.25 = \frac{1}{4} \cdot 5^2$ at $\theta = 90°$, and falls back towards 3.25. A fine scan over θ by computer gives its peak at 90.0°.
- Symmetric case $p = q$, $\theta = 90°$: $ABA'B'$ is itself a square centred at O with side $p\sqrt{2}$, and its midpoint square has side p, area $p^2 = \frac{1}{4}(2p)^2$, as the formula says.
- With random choices of p, q and θ, the midpoints were confirmed by computer to form a square with area $\frac{1}{4}(p^2 + q^2 + 2pq\sin\theta)$.

Common slips

- Taking the angle between OA and OA' to be θ rather than $90° + \theta$. That leads to the area $\frac{1}{4}(p^2 + q^2 - 2pq\cos\theta)$ and the wrong conclusion that θ should approach 180°.
- Showing only that the four sides are equal. That gives a rhombus. You also need a right angle, and here it comes free from $\overrightarrow{QS} = R\,\overrightarrow{PQ}$.

Beyond STEP

A rotation through a right angle is a linear map, $R(x, y) = (y, -x)$, and in the complex plane it is multiplication by $-\mathrm{i}$. The whole argument is a computation with linear maps: $\mathbf{q} - \mathbf{p}$ and $\mathbf{s} - \mathbf{q}$ differ by R because R commutes with subtraction. The same idea gives quick proofs of results such as Napoleon's theorem and van Aubel's theorem. Needham's *Visual Complex Analysis* develops geometry through complex multiplication in exactly this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.